**1. Dataset Overview**

This project uses the **Amazon Reviews Dataset (clean_amazon_reviews_1.csv)** to analyze customer opinions. The dataset contains **34,624 customer reviews** and **10 columns**, including Product Name, Brand, Rating, Review Text, Clean Review, and Sentiment. It is used to identify customer satisfaction and help businesses improve their products and services.

**2. Seller Journey**

The Seller Journey explains how sellers interact with customers on an e-commerce platform.

**Flow:**
Seller Registration → Product Listing → Customer Purchase → Delivery → Customer Review → Sentiment Analysis → Product Improvement

Customer reviews help sellers understand customer satisfaction, improve product quality, make better business decisions, and increase sales.

**3. Existing Systems**

Many online platforms use Sentiment Analysis to understand customer feedback.

- **Amazon:** Improves product recommendations and identifies product issues.
- **Flipkart:** Uses reviews to improve products and customer service.
- **Google Play Store:** Helps developers fix bugs and enhance app features.
- **YouTube:** Analyzes comments to understand audience reactions.
- **Twitter (X):** Monitors public opinion and brand reputation.

**4. Literature Review**

Several studies have shown that Machine Learning and NLP are effective for sentiment analysis. Popular techniques include **TF-IDF**, **Logistic Regression**, **Support Vector Machine (SVM)**, and **LSTM**. These methods help classify customer reviews accurately and provide valuable insights for businesses.

**5. Conclusion**

Sentiment Analysis is an important application of Natural Language Processing (NLP). By analyzing Amazon customer reviews, businesses can understand customer opinions, improve product quality, enhance customer satisfaction, and make informed decisions.

**6. References**


- Bing Liu – *Sentiment Analysis and Opinion Mining*
- Jurafsky & Martin – *Speech and Language Processing*
- Pang & Lee – *Opinion Mining and Sentiment Analysis*
- Google Scholar
- Kaggle Amazon Reviews Dataset


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This version is short enough to fit on **one A4 page** (with normal margins and 11–12 pt font) and is suitable for a college submission.